maximum possible neutron-star mass (curve *A* in Figure 5.3), then a star heavier than that maximum, when it dies, might form a black hole.

Having posed this maximum-mass question with stark clarity, Oppenheimer went about solving it, methodically and unequivocally—and, as was his standard practice, in collaboration with a student, in this case a young man named George Volkoff. The tale of Oppenheimer and Volkoff's quest to learn the masses of neutron stars, and the central contributions of Oppenheimer's Caltech friend Richard Tolman, is told in Box 5.4. It is a tale that illustrates Oppenheimer's mode of research and several of the strategies by which physicists operate, when they understand clearly *some* of the laws that govern the phenomenon they are studying, but *not all:* In this case Oppenheimer understood the laws of quantum mechanics and general relativity, but neither he nor anyone else understood the nuclear force very well.

Despite their poor knowledge of the nuclear force, Oppenheimer and Volkoff were able to show unequivocally (Box 5.4) that *there is a maximum mass for neutron stars, and it lies between about half a solar mass and several solar masses.*

In the 1990s, after fifty years of additional study, we know that Oppenheimer and Volkoff were correct; neutron stars do, indeed, have a maximum allowed mass, and it is now known to lie between 1.5 and 3 solar masses, roughly the same ballpark as their estimate. Moreover, since 1967 hundreds of neutron stars have been observed by astronomers, and the masses of several have been measured with high accuracy. The measured masses are all close to 1.4 Suns; why, we do not know.

Box 5.4

The Tale of Oppenheimer, Volkoff, and Tolman: AQuest for Neutron-Star Masses

When embarking on a complicated analysis, it is helpful to get one's bearings by beginning with a rough, "order-of-magnitude" calculation, a calculation accurate only to within a factor of, say, 10. In keeping with this rule of thumb, Oppenheimer began his assault on the issue of whether